Summary Sheet

Pruned evidence-gated pharmacogenomic LLM-alert capstone

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Conference / Symposium

Field	Value
Name	Machine Learning for Health Symposium (ML4H 2026)
URL	https://ml4h.ahli.cc/
Author instructions / CFP	https://ml4h.ahli.cc/submit/call-for-papers/
Submission deadline	September 10, 2026, 11:59 PM AoE
Author notification	October 22, 2026
Camera-ready deadline	November 7, 2026 (tentative)
Event dates	December 6-7, 2026
Location	Sydney, Australia
Suggested track	Findings Track for preliminary specification-conformance work; Proceedings Track only after independent review.
Editable paper format	LaTeX/Overleaf source: paper/ml4h_findings_evidence_gate.tex with paper/ml4h_findings_refs.bib; compiled PDF: paper/ml4h_findings_evidence_gate.pdf.

Title and Abstract

Title: Evidence-Gated Medical LLM Alerts for Pharmacogenomic Claims: Detecting Overclaiming Relative to Guideline-Supported Evidence

Abstract: LLM-generated medication alerts can sound more certain than their evidence permits. We built a deterministic pre-display monitor for one workflow: pharmacogenomic and genomic medication-alert text. The monitor checks source support, population fit, and claim strength, then returns allow, narrow, abstain, or deny. Stage 1 is a synthetic specification test: the case bank supplies the structured annotations that a future live system would need to extract, and ablation tests whether any check is dispensable. Disabling source support let 2 of 23 designed overclaims pass unchanged; disabling population fit let 1 pass; disabling claim strength let 6 pass. With all checks active, all 23 designed overclaim archetypes were blocked or narrowed and all 10 bounded alerts passed unchanged. Two LLM self-evaluation baselines, GPT-5.6-terra and Grok-4.5, also routed all 23 overclaims, but differed from the deterministic monitor on selected actions or surfaced rationales. The first-triggering check matched the author-assigned check label in 31 of 33 cases, interpreted as precedence sensitivity rather than action failure. These are specification-conformance results, not clinical safety, generalization, or patient-benefit evidence.

Generative AI Use

Generative AI tools were used for brainstorming, critique, code scaffolding, drafting, packaging, formatting, and for an LLM self-evaluation baseline on synthetic annotations only. The baseline used GPT-5.6-terra and Grok-4.5; no real patient records, protected health information, or private genomic data were transmitted. The author selected the final scope, directed the pruning, verified sources, reviewed generated text, ran local tests, and remains responsible for all claims, limitations, and submission materials.

Optional Data, Code, and Tools

- Synthetic 33-case pharmacogenomic/genomic medication-alert case bank with no real patient data, PHI, diagnosis, or treatment recommendation.
- Executable three-check controller with unit tests and deterministic replay outputs.
- One-check ablation CSV, precedence-sensitivity CSV, LLM self-evaluation baseline CSV/JSON outputs, and CSV/JSON summaries for overclaim handling, inappropriate denial, calibration, and check categories.

Evidence-Boundary Framework

The empirical workflow is pharmacogenomics, but the design pattern is broader: source support asks whether evidence exists and applies; population fit asks whether evidence transports to the target patient or population; and claim strength asks whether the evidence supports the displayed action rather than only a weaker association or surrogate endpoint.